

Project-Based Mechanical Design

Apprenticeship Program

Course Objective

This course is designed to develop industry-ready mechanical design engineers by combining Engineering Drawing fundamentals with practical SolidWorks/CATIA modeling, assembly design, and documentation skills. By the end of the program, students will be able to create fully defined 3D models, generate manufacturing-ready drawings, build structured assemblies, and present professional design portfolios suitable for entry-level design engineering roles.

After completing this course, students will be able to:

- Interpret and create engineering drawings using standard practices
- Apply dimensioning, tolerances, and basic GD&T correctly
- Develop parametric 3D part models with proper design intent
- Design and manage mechanical assemblies with correct mates
- Generate Bill of Materials (BOM) and assembly documentation
- Perform basic interference and motion checks
- Create fabrication and sheet metal drawings
- Prepare a professional CAD portfolio for job applications
- Confidently explain design decisions in technical interviews

◆ **MODULE 1 – Engineering Drawing Mastery**

Concepts:

- Orthographic & isometric projections
- First vs Third angle
- Section & detail views
- Line types and standards
- Dimensioning rules (ISO / ASME basics)
- Limits & fits
- Reading industrial drawings

Industry Exercise:

- Interpret 5 real manufacturing drawings
- Reverse engineer drawing → 3D model

Deliverable:

- 5 manually interpreted CAD models
 - 5 professional drawing sheets
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◆ MODULE 2 – Parametric & Feature-Based Modeling

Concepts:

- Design intent
- Fully defined sketches
- Multi-body modeling
- Configurations
- Design tables
- Advanced features (Loft, Sweep, Shell, Draft)
- Surface modeling basics

Industry Integration:

- Model parts based on manufacturing constraints
- Redesign flawed geometry

Projects:

1. Parametric mounting bracket
2. Flange with bolt pattern
3. Shaft with bearing seat
4. Injection molded housing

Deliverables:

- 4 parts
 - 4 detailed drawings
 - Tolerance applied
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◆ MODULE 3 – GD&T + Tolerance Engineering

Concepts:

- GD&T symbols & interpretation
- Datums & feature control frame
- Position tolerance
- Flatness, Parallelism, Perpendicularity
- Fit calculation
- Tolerance stack-up basics

Industry Task:

- Apply GD&T on shaft-bearing system
- Perform basic tolerance stack-up

Deliverable:

- 2 GD&T applied drawings
- Stack-up calculation sheet

◆ MODULE 4 – Sheet Metal, Weldments & Fabrication

Concepts:

- Bend allowance
- K-factor
- Flat pattern export
- Weld symbols
- Structural frames
- Cut list documentation

Project:

Design industrial electrical enclosure

Deliver:

- 3D model
 - Flat pattern
 - Fabrication drawing
 - Weld symbol placement
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◆ **MODULE 5 – Assembly Design & BOM Strategy**

Concepts:

- Bottom-up & Top-down modeling
- Mates & advanced mates
- Subassemblies
- Standard component library
- Interference detection
- Exploded view
- Bill of Materials

Industry Project:

Two-stage Industrial Gearbox

Deliver:

- 3D assembly
 - Section view drawing
 - BOM
 - Exploded animation
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◆ **Module 6 – Assembly Design**

1) Assembly Basics

- Bottom-up vs Top-down approach
- Creating assemblies
- Inserting components
- Understanding degrees of freedom

2) Mates & Constraints

- Standard mates (Coincident, Concentric, Parallel)
- Advanced mates (Limit, Width)
- Fully defining an assembly
- Avoiding over-constraining

3) Sub-Assemblies

- Creating sub-assemblies
- Assembly structure management
- Organizing feature tree properly

4) Interference Check

- Interference detection
- Clearance verification
- Fixing assembly errors

5) Bill of Materials (BOM)

- Generating BOM
- Adding balloons
- Exploded view creation
- Assembly drawing basics

6) Basic Motion Study

- Applying motor
- Simple mechanism motion
- Animation export

◆ MODULE 7 – Capstone Industry Project

Choose ONE:

Option A – Automotive Suspension Subassembly

Option B – Industrial Hydraulic Press

Option C – Electric Go-Kart Frame

Must Include:

- Complete CAD modeling
- Assembly structure
- GD&T
- Manufacturing drawings
- BOM
- Basic static simulation
- Design validation note
- 15-slide technical presentation

Evaluation System

- Weekly graded assignments
- Drawing audit feedback
- Design review simulation
- Viva session
- Final project defense

What Learners Will Have at the End of the Program

By the completion of this program, students will possess both **technical skills** and a **professional portfolio** ready for job applications.

Portfolio & Technical Deliverables

- 10+ fully defined 3D part models
- 3 complete mechanical assemblies
- 5+ manufacturing-ready engineering drawings
- 2 fabrication/sheet metal drawings
- 1 major capstone project (industry-style)
- Structured Bill of Materials (BOM)
- Exploded view assembly drawing
- Section assembly drawing
- Basic motion study animation
- GD&T applied drawing set
- Simulation report (basic structural validation)
- Final technical presentation deck